Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Binomial Expansion

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Expand $(1 + x)^4$. **[3]**

2. Expand $(1 + 2x)^3$. **[3]**

3. Expand $(1 - x)^5$. **[3]**

4. Expand $(2 + x)^4$. **[4]**

5. Expand $(3 - x)^3$. **[3]**

6. Expand $(1 + 3x)^4$. **[3]**

7. Expand $(2 - 3x)^3$. **[4]**

8. Find the coefficient of x^2 in $(1 + x)^6$. **[2]**

9. Find the coefficient of x^3 in $(1 + 2x)^5$. **[3]**

10. Find the coefficient of x^2 in $(1 - 3x)^4$. **[3]**

11. Find the first three terms of $(1 + x)^7$. **[3]**

12. Find the first three terms of $(1-2x)^6$. **[3]**

13. Expand $(x+2)^4$. **[3]**

14. Expand $(2x-1)^4$. **[4]**

15. Find the constant term in $\left(x+\frac{1}{x}\right)^4$. **[3]**

Section B – Hard (Q16–25)

16. Find the coefficient of x^3 in $(1+x)^5(1-x)^3$. **[5]**

17. Find the coefficient of x^2 in $(2+x)^4(1-x)^3$. **[5]**

18. Find the coefficient of x in $(1+2x)^4(1-x)^2$. **[5]**

19. Expand $(1+x+x^2)^3$ up to x^2. **[5]**

20. Find the term independent of x in $\left(x^2+\frac{2}{x}\right)^6$. **[5]**

21. Find the term independent of x in $\left(2x-\frac{1}{x}\right)^5$. **[5]**

22. Use the binomial expansion to approximate 1.01^5 to 4 d.p. **[4]**

23. Use the binomial expansion to approximate 0.99^4 to 4 d.p. **[4]**

24. Find n if the coefficient of x^2 in $(1+x)^n$ is 45. **[4]**

25. Find n if the coefficient of x^3 in $(1+2x)^n$ is 160. **[5]**

End of Worksheet • *Show all working clearly.*